AP CALCULUS AB

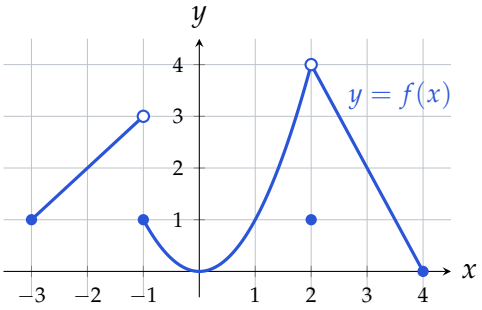
Unit 1 • Self Check 1.1 • Lessons 1-1 and 1-2

Self Check 1.1 • Lessons 1-1 and 1-2

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



Question 1

NO CALCULATOR

Using the graph of f, $\lim_{x \to -1^+} f(x) =$

(A) 3
(B) 1
(C) 0
(D) It does not exist.

Question 2

NO CALCULATOR

Using the graph of f, $\lim_{x \to 3} f(x) =$

(A) 1
(B) 3
(C) 2
(D) 4

Question 3 NO CALCULATOR

$\lim_{x \to -2} \frac{x^2 - 4}{x + 2} =$

(A) 0
(B) -4
(C) 4
(D) It does not exist.

Question 4 NO CALCULATOR

$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4} =$

(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) 0
(D) 4

Question 5 (Free response) NO CALCULATOR

Find $\lim_{x \to 0} \frac{x^2 + 5x}{x}$, $\lim_{x \to 1} \frac{\frac{1}{x} - 1}{x - 1}$ and $\lim_{x \to 0} \frac{\sin 4x}{2x}$.